Table 3: Industry usage of quantum computing

Industry	Usage
Health care and pharma services	• Accelerating drug discovery • Drug design, molecular dynamics • Faster identification of candidate compounds and novel materials design • Optimising therapy treatment • Time to market of new drugs
Financial services	• Trading strategies • Portfolio optimisation • Asset pricing and risk analysis • Detecting market instabilities
Transportation	• Self-driving cars • Air traffic management • Managing transportation requests • Goods delivery in a short time
Energy	• Network design • Energy distribution • Oil well optimisation
Governments	• Quantum-safe encryption • National security and defence • Cryptography and cybersecurity • Secure communications • Supply chain resilience, logistics planning, and resource allocation across military and civilian agencies

find out the best possible decision out of the large number of decisions or options that are available
- Search, sampling and pattern matching
- Quantum encryption

The challenges

Lack of proven software advantage: Only a handful of quantum algorithms (VQE, QAOA, Grover, Shor) demonstrate meaningful, proven speedups today; most real-world problems still lack demonstrated quantum advantage on hardware.

Hybrid orchestration complexity: Integrating quantum subroutines into classical workflows requires robust middleware, cross-stack compatibility, and new programming paradigms.

Technological/error-correction overhead: Even with 2025–2026's error-correction breakthroughs, encoding one logical qubit still requires dozens to hundreds of physical qubits, depending on the code family. Detecting, controlling, and correcting errors in real time remains a major engineering challenge.

Benchmarking and standardisation gap: The absence of industrywide metrics and transparent performance benchmarks hampers an objective comparison of hardware and software stacks.

Vendor lock-in: Proprietary SDKs and cloud APIs lack unified standards, complicated multi-platform development and collaboration.

Energy and infrastructure costs at scale: Quantum computing at scale requires massive physical infrastructure like cryogenics, control electronics, specialised materials, and high-density wiring that drive energy consumption to levels comparable to or exceeding modern data centres. 'Quantum' does not inherently mean 'low-energy', and scaling towards fault-tolerance introduces steep thermodynamic and engineering overheads.

Paucity of talent and academia-industry collaboration: Demand for quantum-literate engineers, physicists, and algorithm designers continues to outstrip supply, and closer collaboration between industry and academia remains a persistent need.

Cryogenic infrastructure challenges: Quantum computers operate at temperatures close to absolute zero, colder than the vacuum of space. Maintaining such a low temperature is a big challenge.

Accelerating post-quantum cryptography migration: The global shift to post-quantum cryptography (PQC) is now time-bound. With formal PQC standards finalised and government compliance deadlines converging in the early 2030s, enterprises face a shrinking window to inventory their cryptographic assets, modernise protocols, and mitigate 'harvest-now, decrypt-later' risks before they translate into material breaches.

Modern enterprises are actively exploring cloud-based quantum computing to enhance productivity and operational efficiency, anticipating that quantum computing, combined with advanced AI, machine learning, and business analytics, will deliver greater agility in their business processes. Enterprises are also increasingly exploring quantum hardware roadmaps, error-mitigation techniques, and hybrid quantum-classical algorithm frameworks that bridge classical systems with emerging quantum processors, while, in parallel, urgently building post-quantum cryptography readiness ahead of the eventual arrival of a cryptographically relevant quantum computer. END

By: Dr Gopala Krishna Behara and Narumanchi Sree Keerthi

Dr Gopala Krishna Behara is an enterprise architect with nearly three decades of experience in the IT Industry.

Narumanchi Sree Keerthi is a senior software engineer with around 5 years of IT experience.